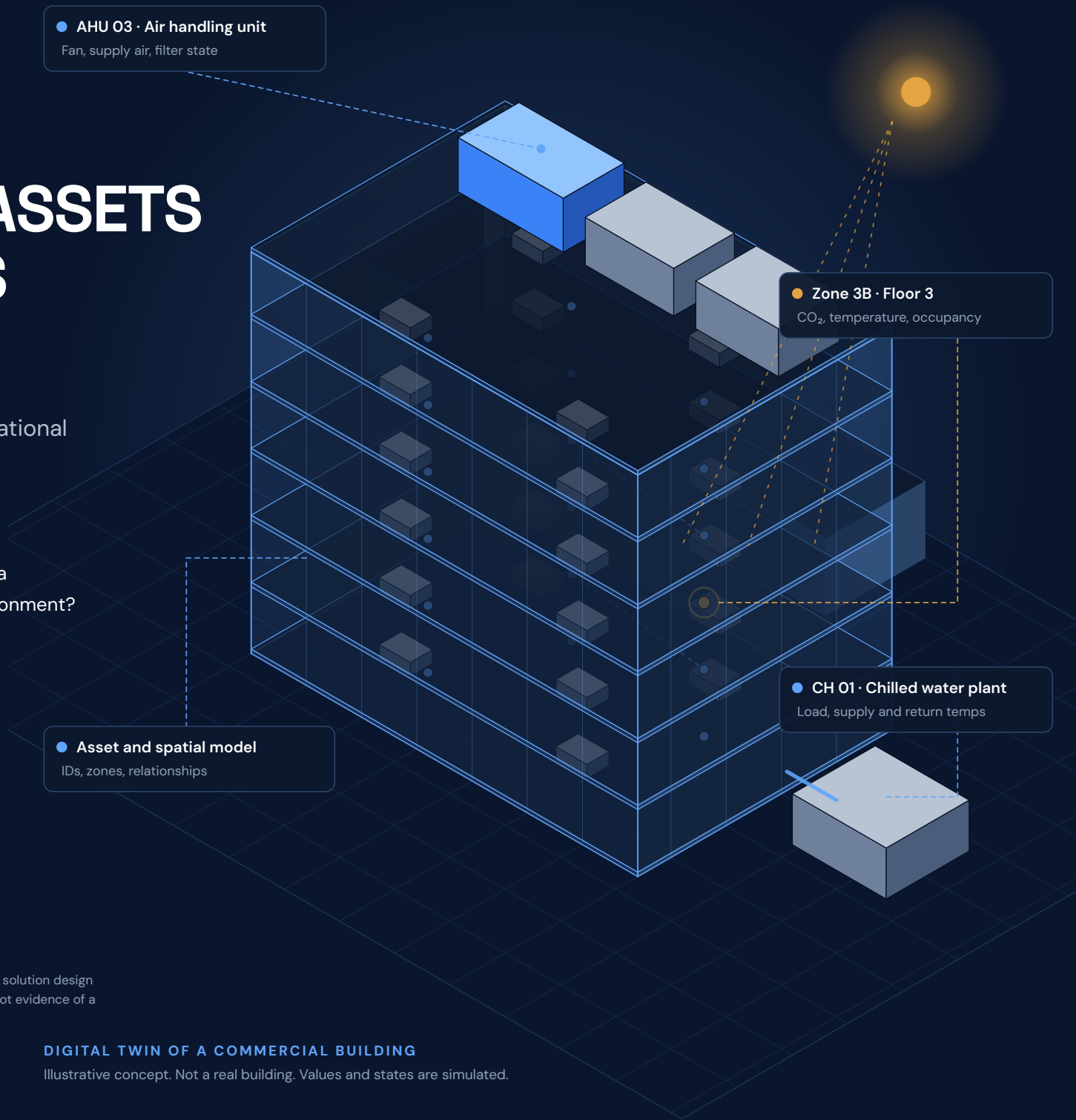
INDEPENDENT TECHNICAL CASE STUDY

FROM PHYSICAL ASSETS TO AUTONOMOUS OPERATIONS

Designing the Information, Integration, and Operational Workflows Behind Intelligent Buildings

THE CENTRAL QUESTION

What does it take to transform a physical building into a trustworthy, context aware, intelligent operational environment?



DIGITAL TWIN OF A COMMERCIAL BUILDING

Illustrative concept. Not a real building. Values and states are simulated.

Mark Jerome Gomez

Product Manager and Licensed Electronics Engineer
IoT, smart infrastructure and digital twin platforms

INDEPENDENT TECHNICAL CASE STUDY | NOVEMBER 2026

A personal conceptual case study based on publicly available information and general solution design principles. All proposed capabilities and interface illustrations are illustrative and are not evidence of a deployed solution.

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THE STRATEGIC OPPORTUNITY

Intelligent Buildings Depend on Connected, Trustworthy Physical Context

A commercial building already holds most of the data an intelligent platform needs. The difficulty is that the data sits in separate systems that were procured, commissioned and named independently: HVAC plant, the building management system, energy metering, occupancy, environmental sensors, asset management, and security and operations platforms.

WHAT FRAGMENTATION COSTS

- **Limited cross system visibility.** Each team sees one slice of building behavior.
- **Reactive investigation.** Incidents are explained afterward, by hand, across several screens.
- **Hidden dependencies.** It is hard to see which zones an air handler serves or what a chiller fault will touch.
- **Fragmented workflows.** Alarms, work orders and energy reviews rarely share context.
- **Underused data.** Telemetry is stored but seldom tied to spatial and asset meaning.

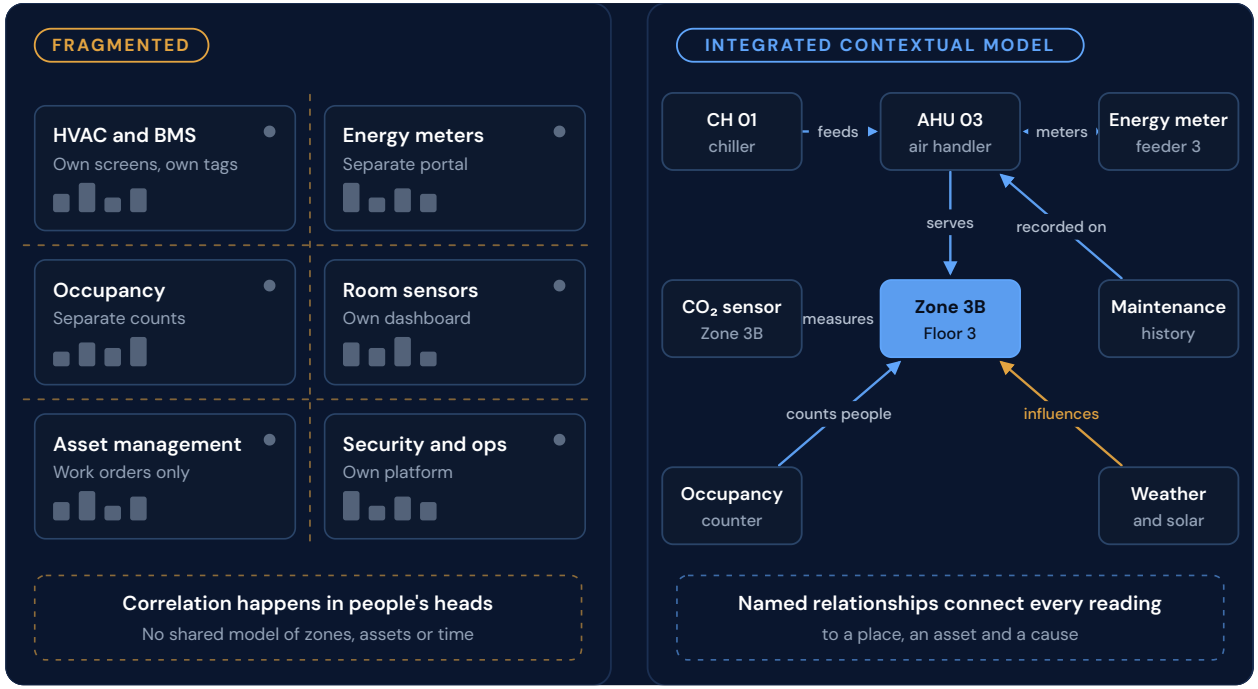


Figure 1. Fragmented systems (left) versus an integrated contextual model (right). Relationships are named and typed rather than implied. Entities shown are illustrative.

AUTHOR ANALYSIS This study starts one step before optimization: what it takes to make an existing estate legible enough to optimize safely.

HOW TO READ THIS DOCUMENT

INDUSTRY PRACTICE established and publicly documented. AUTHOR ANALYSIS my own conceptual reasoning. ILLUSTRATIVE simulated or conceptual, not a deployed solution.

FIVE CHALLENGES THAT STRUCTURE THIS STUDY

01
Physical to digital fidelity
Assets, BIM and identifiers become a reliable digital record.
PAGE 3

02
Connected infrastructure
BMS, IoT and gateways interoperate without full replacement.
PAGE 4

03
Contextual intelligence
Occupancy, weather and equipment behavior are understood together.
PAGE 5

04
Decision and action
Recommendation, authorization and closed loop verification.
PAGES 6 AND 7

05
Productization and scale
Commissioning and validation that repeat across buildings.
PAGE 8

CHALLENGE 1 | PHYSICAL TO DIGITAL FIDELITY

From BIM and MEP Information to Operational Asset Intelligence

Before a platform can reason about a building, every important asset must exist digitally with a stable identity, a location, relationships and live data. That is an information problem before it is an AI problem.

SEVEN ELEMENTS OF THE FOUNDATION

- **Asset identity.** One identifier shared by the model, the BMS point list and the maintenance record.
- **BIM and IFC information.** Geometry plus classified, parameterized asset data; IFC is the open exchange schema.[1,2]
- **MEP relationships.** Which plant feeds which air handler, and which air handler serves which terminal units.
- **Spatial hierarchy.** Site, building, floor, zone, room.
- **Telemetry mapping.** Each point tied to the asset it describes, with units and meaning.
- **Equipment to zone association.** The link that turns a reading into an operational impact.
- **Validation and commissioning.** As built checks, point to point tests and data quality rules before anything is trusted.

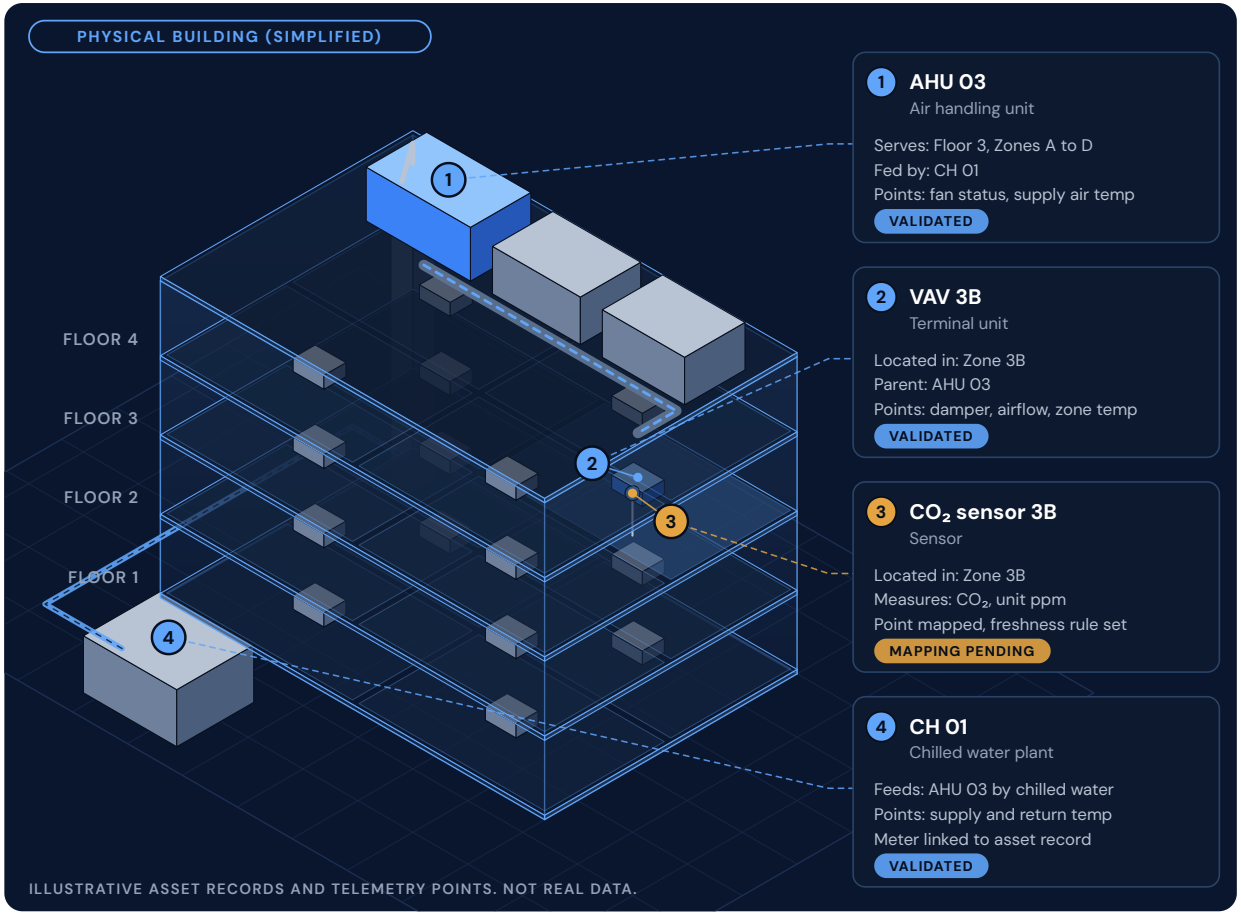


Figure 2. Annotated cutaway linking physical equipment, digital asset records and operational telemetry. Records, states and values are illustrative.

Accurate information is a prerequisite for trustworthy intelligence. A model that is visually right but informationally wrong produces confident, wrong answers.

AUTHOR ANALYSIS

INDUSTRY PRACTICE ISO 19650 supplies the information management discipline: requirements, naming conventions and a common data environment.[1]

AUTHOR ANALYSIS A high detail MEP model is the foundation for digital twins, predictive maintenance and any later automation. This page views that dependency from the operations side.

CHALLENGE 2 | CONNECTED INFRASTRUCTURE

Making Heterogeneous Building Systems Work Together

Existing buildings speak many dialects. Integration means translating them into one consistent, trustworthy stream without asking the owner to replace what already works.

WHAT INTEGRATION INVOLVES

- **BMS integration.** The supervisory system is usually the richest source of plant data and the gatekeeper for any control action.
- **Protocols.** BACnet is common in building automation, Modbus in meters and plant equipment, MQTT for lightweight device telemetry; KNX appears in some room level systems.[3,4,5]
- **Gateways and controllers.** Translate field protocols and keep OT networks separated from IT.[7]
- **Normalization.** Consistent names, units, sampling and meaning, tied to the asset model.
- **Freshness and quality.** Every value carries its age, source and a confidence flag.
- **Existing building realities.** Undocumented points, inconsistent naming, mixed firmware, intermittent links and limited write access.

Read before write. Observation earns the right to recommend; recommendation earns the right to act.

AUTHOR ANALYSIS

AUTHOR ANALYSIS A hardware agnostic approach, working with legacy systems instead of replacing them, is the realistic path for most existing buildings.

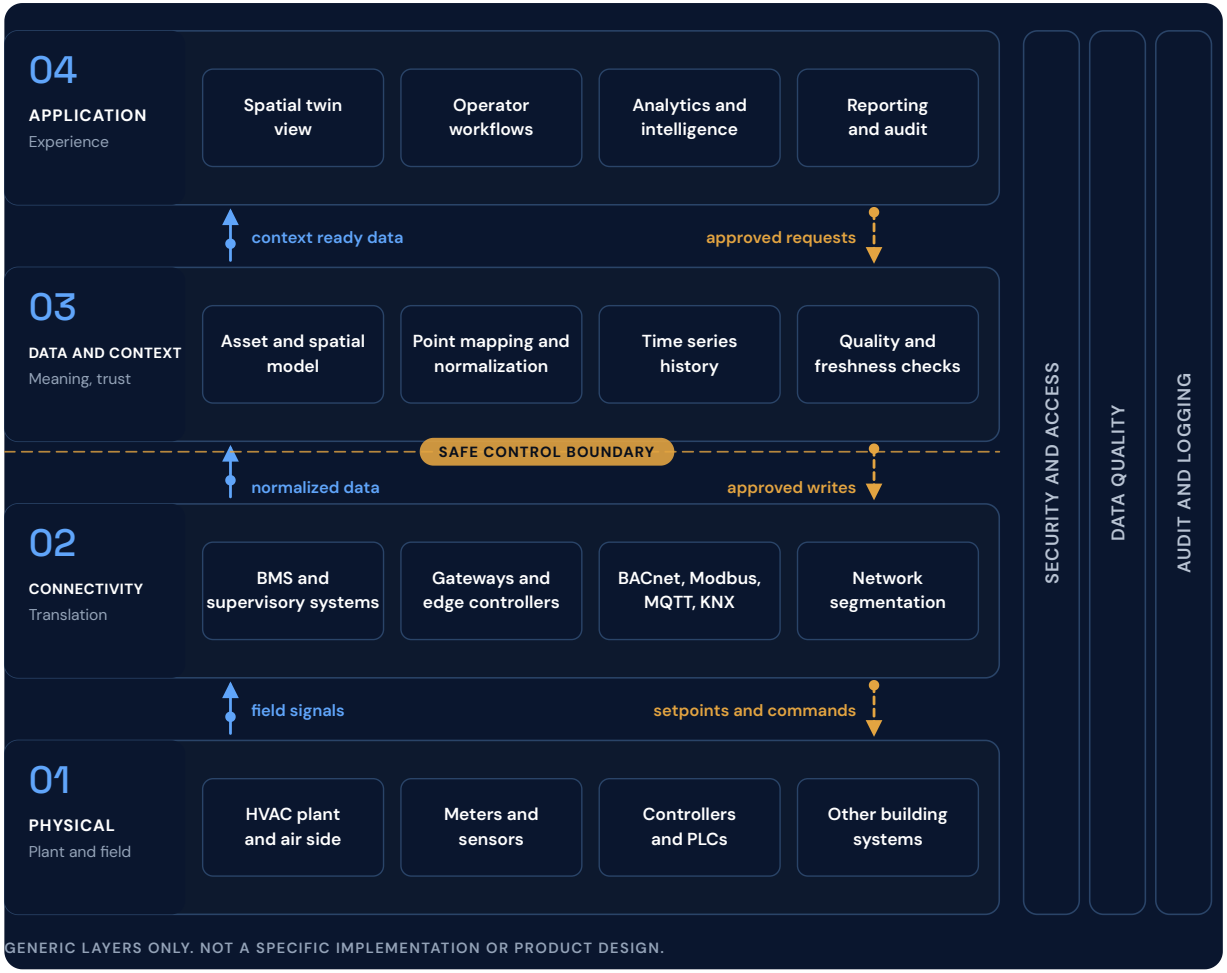


Figure 3. High level integration layers with a safe control boundary. Component names are generic categories, not a specific design or product.

CHALLENGE 3 | CONTEXTUAL INTELLIGENCE

Understanding Building Behavior in Context

A reading means little alone. A CO_2 value of 1,180 ppm is a number. The same value in a crowded, west facing zone on a hot afternoon, served by an air handler that is lagging, is a situation.

ILLUSTRATIVE SCENARIO SIMULATED

1. Outdoor temperature rises through the afternoon.
2. Solar exposure builds on the west façade.
3. Occupancy increases on Floor 3.
4. Indoor CO_2 rises in Zone 3B.
5. HVAC demand increases at AHU 03.
6. AHU 03 behaves abnormally against its own history.

With spatial and asset relationships in place, a platform can connect these six events to the affected zone and its serving equipment, instead of raising six unrelated alarms. Solar gain, occupant heat and ventilation load all end up as demand on the same plant, so reasoning that respects those physical constraints is more useful than pattern matching alone. AUTHOR ANALYSIS

> INDUSTRY PRACTICE CO_2 is widely used as an indicator of ventilation relative to occupancy.[8]

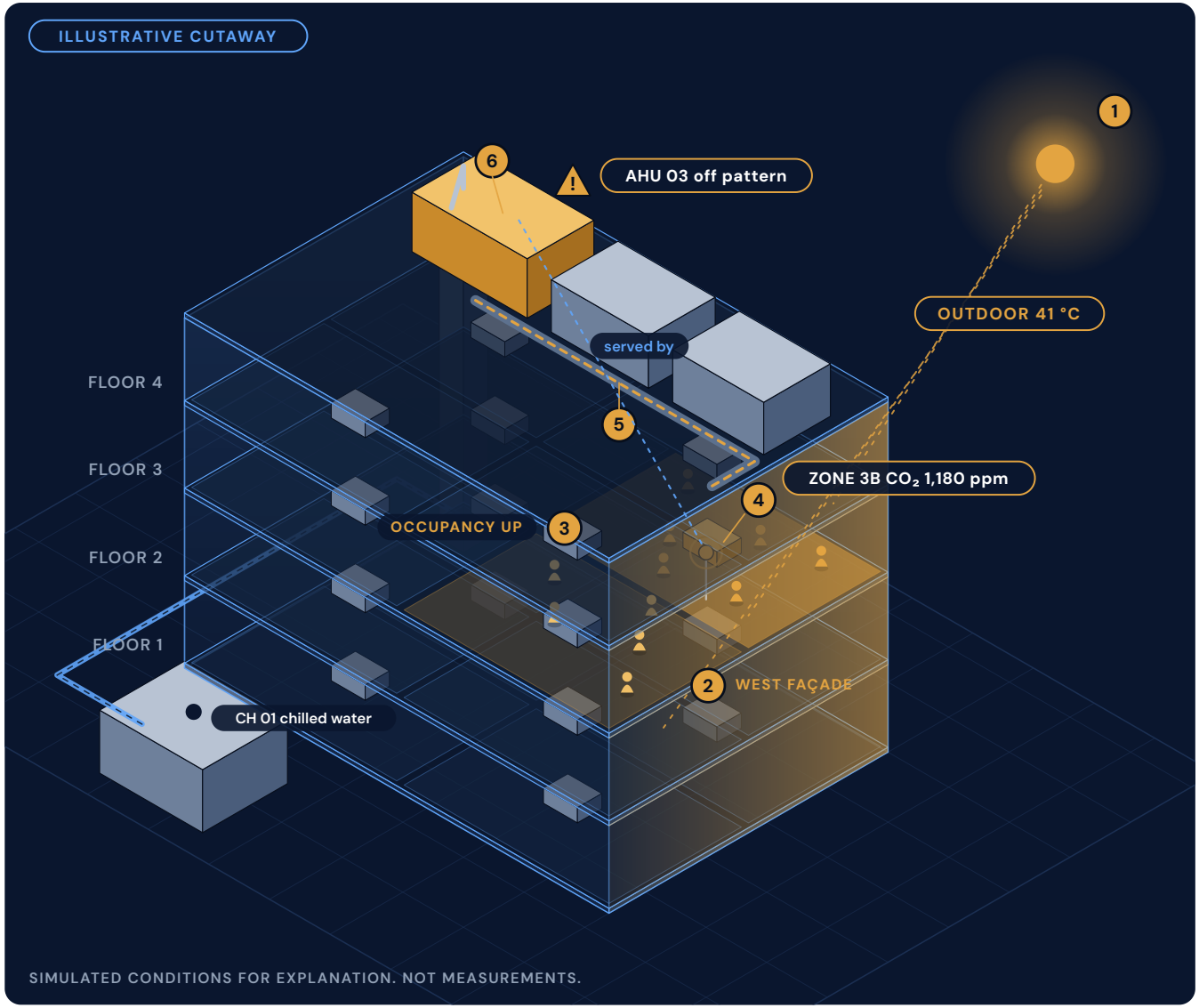


Figure 4. Illustrative cutaway with simulated conditions. Occupancy, temperatures, CO_2 and equipment states are invented for explanation and are not measurements.

CHALLENGE 4 | DECISION AND ACTION

A Closed Loop Approach to Intelligent Building Operations

HUMAN GATE

WITHIN LIMITS

1 PERCEIVE	2 CONTEXTUALIZE	3 EVALUATE	4 RECOMMEND	5 AUTHORIZE	6 ACT	7 VERIFY
Collect validated telemetry and equipment state.	Place readings in zone, asset and weather context.	Compare with history and expected behavior.	Weigh candidate interventions and trade offs.	An accountable person approves within set limits.	Execute through the BMS inside a control boundary.	Check whether the outcome matched the intent.

Outcome and context feed the next cycle

SITUATION	A high occupancy zone shows rising CO_2, higher cooling demand and abnormal equipment behavior.
INTERPRETATION	The platform correlates conditions, equipment state, spatial relationships and history.
DECISION SUPPORT	Candidate interventions are evaluated against operating constraints, with trade offs explained.
SUPERVISION	An authorized operator reviews, modifies or approves. Nothing executes without that step.
VERIFICATION	Later conditions are compared with the intended outcome and recorded for learning.

AUTHOR ANALYSIS **Safe control boundary:** a short list of permitted points, limited ranges, rate limits, fallback to existing BMS logic and a full audit trail.

AUTHOR ANALYSIS Systems that sense, reason, decide and act need a feedback loop to improve. The authorization step and control boundary here are my proposed considerations for how such autonomy earns trust.

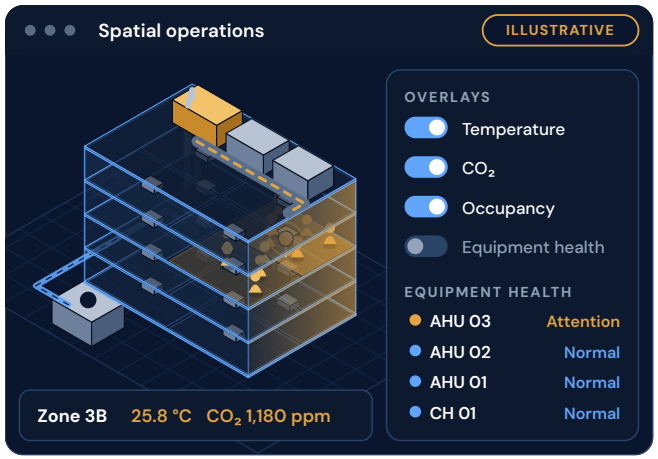


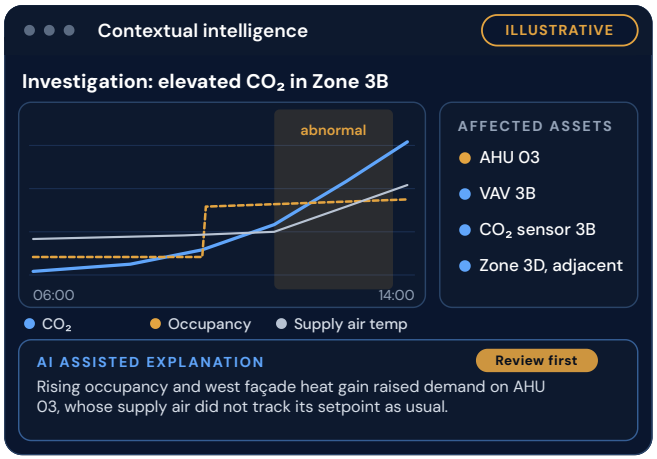
Figure 5. Conceptual decision support interface. **Conceptual UI, Illustrative Only.** Values are simulated; this is not a screenshot of a working application.

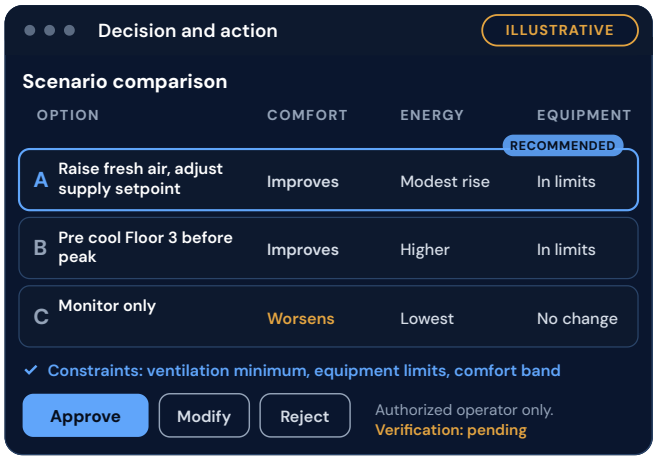
CHALLENGE 4 CONTINUED | PRODUCT EXPERIENCE

Translating Technical Intelligence into Usable Building Operations

Each view answers one operational question for one kind of user. The aim is to show workflow design judgment, not to suggest that any existing product lacks these capabilities.







A Spatial Operations View

USER	Facility or operations manager.
QUESTION	Where is something changing, and which equipment is involved?
SHOWS	Building view, zone conditions, equipment health, contextual overlays.
ACTION	Select a zone, toggle overlays, open an investigation.
BENEFIT	Faster orientation with less hunting across screens.

B Operational Intelligence View

USER	Operations engineer or BMS specialist.
QUESTION	What is happening, and what could be causing it?
SHOWS	Incident context, affected assets, historical trends, possible causes with evidence.
ACTION	Review evidence, then confirm or dismiss a possible cause.
BENEFIT	Investigation starts from context instead of a blank search.

C Decision and Verification View

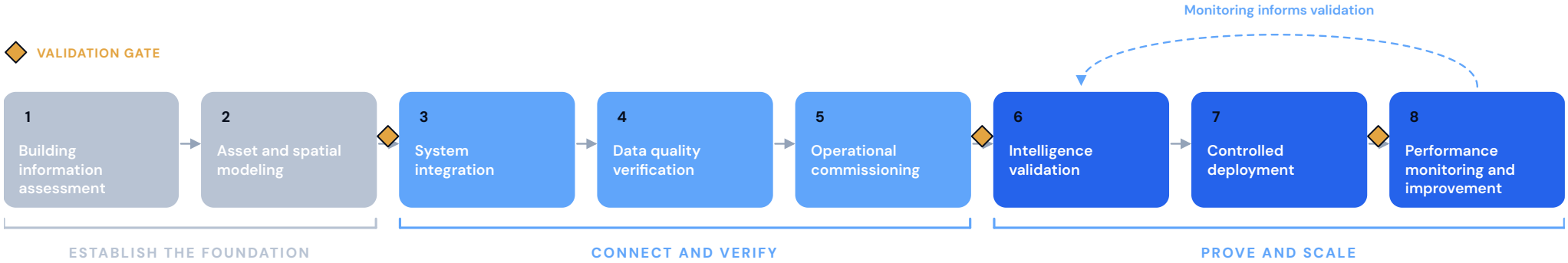
USER	Authorized operator or building manager.
QUESTION	What should we do, and did it work?
SHOWS	Option comparison, constraints, expected effect, observed result.
ACTION	Approve, modify or reject; review the verification.
BENEFIT	Decisions that are explicit, explainable and auditable.

Figure 6. Three conceptual views. **Conceptual UI, Illustrative Only.** They are not screenshots of a working application and are not modeled on any existing product interface. AUTHOR ANALYSIS

CHALLENGE 5 | PRODUCTIZATION AND SCALE

From a Digital Model to a Trusted Operational System

Scale comes from a repeatable method: the same lifecycle applied to every building, with evidence gathered at each gate. This is a conceptual lifecycle. It carries no timelines and no performance claims.



Establish the foundation

Stages 1 and 2

KEY DEPENDENCIES

Access to drawings, BIM or survey data; naming and classification standards; owner decisions on scope.

RISKS TO MANAGE

Outdated or missing as built information; inconsistent asset naming across sources.

VALIDATION

Model to site spot checks and reconciliation of the asset register.

Connect and verify

Stages 3 to 5

KEY DEPENDENCIES

BMS access and permissions; network and cybersecurity approval; gateway availability.

RISKS TO MANAGE

Undocumented points, unstable links, vendor access limits and commissioning effort in occupied buildings.

VALIDATION

Point to point tests, freshness and completeness rules, and operator sign off.

Prove and scale

Stages 6 to 8

KEY DEPENDENCIES

A baseline period, agreed success measures and clear authority for any control action.

RISKS TO MANAGE

Plausible but wrong recommendations, limited operator trust, and drift as buildings change.

VALIDATION

Back testing on history, supervised trials, before and after comparison with an agreed method, and audit review.

AUTHOR ANALYSIS Where value is measured against outcomes, a measurement method that every party trusts is central to this lifecycle. Rollout building by building depends on it.

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PROFESSIONAL ALIGNMENT

Connecting Engineering, Product Thinking, and Real World Operations

This study reflects how I approach intelligent building problems: start from physical truth, respect the systems that already exist, and design for the people who must trust the result. My roles have exposed me to the interconnected disciplines it connects.

ALDAR Properties

OCT 2024 TO OCT 2025

Product Owner and Smart City Lead

Smart city and connected infrastructure initiatives, including building platforms at Saadiyat Grove, digital twin related integrations, smart parking and command and control environments.

TERMINUS Group

OCT 2025 TO AUG 2026

Product Manager, Enterprise AI and IoT Platforms

AIoT and smart city platform strategy, connected product requirements, solution evaluation and predictive maintenance use cases.

Disrupt-X

MAR 2023 TO OCT 2024

Product Engineer, then Product Manager

IoT platforms, connected devices, telemetry and platform workflows for facility management.

Foundation: BS Electronics Engineering, Licensed Electronics Engineer, and MBA. Roles, titles and dates follow my professional record.

> **Scope of this document.** It describes concepts and judgment. It does not describe deliverables, ownership, results or architecture from those roles, nor any vendor's product.

PROFESSIONAL PERSPECTIVE

"The value of a digital twin is not simply in representing physical infrastructure digitally, but in enabling people to understand how interconnected systems behave, evaluate operational decisions, and act with greater confidence."

Portfolio: https://mark-gomez.vercel.app/

DISCIPLINES THIS STUDY CONNECTS

Engineering fundamentals · Connected infrastructure · Systems integration · Product requirements · Operational workflows · Digital experience · Technical delivery

CAPABILITY DEMONSTRATED, NOT CLAIMED AS EXPERIENCE

- **Contextual twin principles** and physics aware reasoning.
- **Cross system IoT, OT and BMS** integration thinking.
- **Asset information quality** and spatial modeling.
- **Decision workflows,** control boundaries and verification.
- **Commissioning and scale** across a portfolio of buildings.

SOURCES

1 ISO 19650 series, information management using building information modelling.
2 buildingSMART International, Industry Foundation Classes (IFC), ISO 16739.
3 ANSI/ASHRAE Standard 135, BACnet.
4 Modbus Organization, Modbus Application Protocol Specification.
5 OASIS, MQTT Version 5.0.
6 ISO/IEC 30173:2023, Digital twin: concepts and terminology.
7 IEC 62443 series, security for industrial automation and control systems.
8 ANSI/ASHRAE Standard 62.1, ventilation and acceptable indoor air quality.

Disclaimer. A personal conceptual case study based on publicly available information and general solution design principles. All proposed capabilities and interface illustrations are illustrative and are not evidence of a deployed solution.

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